

Bridging Correlation and Causation: Recovering Causal Structure in Psychological Networks

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1 Introduction

Psychological network models represent symptoms as nodes and their associations as edges, while the theories that motivate them concern something different, the processes by which one symptom brings about another are seldom well backed. The gap is crucial in order to establish a true causal relationship and prevent flawed assumptions. This project asks how much causal structure the cross-sectional datasets collected in ReBayesed can actually determine, and what - in the data or in the methods - sets the limit on the orientation of an edge.

Research problem. In these networks an edge is a partial correlation: an association that survives conditioning on (i.e., holding constant) every other measured symptom (Borsboom and Cramer, 2013). Causal discovery narrows the possible orientations of such edges using the pattern of conditional independencies, but observational data typically exhibits several structures compatible with the same pattern. There is still yet to be a systematic assessment across a published corpus of psychological networks regarding the extent of the ambiguity.

Relevance. Undirected networks are routinely used to discuss intervention, yet, ironically, intervention requires direction. The ReBayesed export shows how thin the evidence often is for the undirected edges: in the tenth of networks with the least data per parameter, 56.7% of edges have inconclusive Bayesian evidence, against 17.1% in the tenth with the most. If an edge cannot be established as present, its direction is unlikely to fare better, and I would like to test whether that expectation holds.

Significance. Of the 294 networks, 147 have fewer than 5 observations per edge parameter and 44 have fewer than 1. If it turns out that the orientation is simply a consequence of the volume of the data, the findings could be used as guidance for sample planning in future studies. On the other hand, if the orientation stays ambiguous despite presence of strong edge evidence, the findings would identify a limit that more data would not remove. This would be a win-win scenario in either case.

Novelty. Huth et al. (2026) quantified the evidence for undirected edges; Huber (2024) set out why causal structure is often not identifiable from observational data. Neither applied the other's question to this corpus on studies in ReBayesed, and I would like to cross examine the networks using their theories.

2 Related work

Huth et al. (2026) re-analysed 294 cross-sectional psychological networks from 126 papers with Bayesian model averaging and showed that edge evidence depends strongly on sample size. Their export is the input here, because it lets me ask of the same datasets whether the evidence that supports an edge also supports a direction.

Huber (2024) explains why observational data can leave causal structures indistinguishable, particularly when common causes are unmeasured. I use this framework to fix the assumptions under which any orientation the analysis produces should be read.

For the analysis I will use `causal-learn` (Zheng et al., 2024), which implements both PC and FCI in Python. PC assumes that all common causes are measured, and FCI drops that assumption at the cost of weaker conclusions. Using both algorithms from the same library reduces implementation differences as a potential source of variation, making it easier to interpret differences between PC and FCI in light of their different assumptions.

It is however, certainly possible that the algorithms themselves occasionally lead to different results when run repeatedly on larger networks due to different starting positions and local maxima.

3 Research questions

Main question: how much causal structure can be recovered from psychological datasets (that are part of the ReBayesed project), and what limits that recovery?

1. Do networks with more observations per edge parameter yield a greater proportion of determined edge orientations?
2. Does stronger Bayesian evidence for an edge translate to a determined direction for that edge?
3. How do the conclusions change when unmeasured common causes are allowed?

As a supplementary comparison, I will check whether independent studies that measured the same variables agree on orientation.

4 Expected outcomes

Approach. The export contains correlation matrices, sample sizes and metadata rather than participant-level observations, so the first task is to confirm that Fisher’s z tests, and `causal-learn` on top of them, can run from a correlation matrix and a sample size alone. The export would also need to be cleaned before anything runs on it. 233 of the 294 networks have variable names that disagree with their matrix labels, and the export holds 294 networks where the article reports 293. The data comprises of 71 continuous, 196 ordinal and 27 mixed networks. This was discovered during an exploratory data analysis.

RQ1: To what extent is the number of observations per edge parameter associated with the proportion of edges that can be oriented? I will run PC on each network in the subset, record its adjacencies and the proportion of edges it orients, and then check how sensitive those proportions are to the significance threshold and to variable ordering. Because 58 of the 126 papers report several networks that are potentially dependent, I will select one network per

paper by a prespecified rule and report the others separately, so that dependent networks do not inflate the trend. For papers containing multiple eligible networks, I will randomly select one network using a fixed seed for the primary analysis and analyse the remaining networks separately as a sensitivity check. I expect networks with more observations per edge parameter to have a higher proportion of oriented edges, but I do not expect all edges to become oriented as data volume increases.

RQ2: To what extent is Bayesian evidence for edge presence associated with the proportion of edges that can be oriented? I will explore whether networks with stronger Bayesian evidence for their edges also tend to have a greater proportion of oriented edges. These two measures capture different properties of a network: strong evidence that an edge is present does not necessarily provide the conditional-independence information required to determine its direction. The Bayesian edge probabilities are also estimates based on a stochastic search over a subset of possible networks. Therefore, I will treat this as an exploratory comparison rather than assuming that stronger edge evidence should lead to more determined orientations.

RQ3: To what extent does allowing for unmeasured common causes change the causal structure that can be recovered? I will run FCI on the same subset and compare its results with PC at two levels. First, I will use descriptive measures such as the total number of oriented edges and agreement in direction where both algorithms provide an orientation. Second, I will compare more complex structural conclusions, such as ancestral relationships, to examine how allowing for unmeasured common causes changes what can be inferred from the networks. Both algorithms still rely on assumptions such as acyclicity, the Markov and faithfulness conditions, which limit how their results can be interpreted.

Cross-study comparison and outputs. Depression is the largest subtopic, with 32 networks, and the first candidate for comparing orientations across studies; whether that comparison is feasible depends on the metadata cleaning in week 4. Where studies disagree, I will distinguish a direction left unresolved from one that conflicts across studies, since only the second bears on replication. The deliverables are a cleaned pipeline, reproducible analyses, figures and a written interpretation that claims no more than the data determine under the assumptions above.

5 Research plan

The project runs for 13 weeks and is now in week 5. The first three weeks went to exploratory analysis of the export. The plan below is a roadmap of my timely approach to the research.

Week	Activity and output
4	EDA and metadata cleaning, identify eligible networks and studies.
5	Validate Fisher's z and <code>causal-learn</code> on correlation-matrix input.
6	Build and validate the PC pipeline, and run PC on the subset.
7	Summarise adjacencies and orientations, and analyse data volume and sensitivity (RQ1).
8	Relate orientation to Bayesian evidence (RQ2).
9	Run FCI on the matched subset.
10	Compare analyses with and without unmeasured causes (RQ3).
11	Complete the cross-study comparison and checks.
12	Complete the thesis draft and discuss feedback.
13	Revise, check reproducibility and prepare the presentation.

References

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